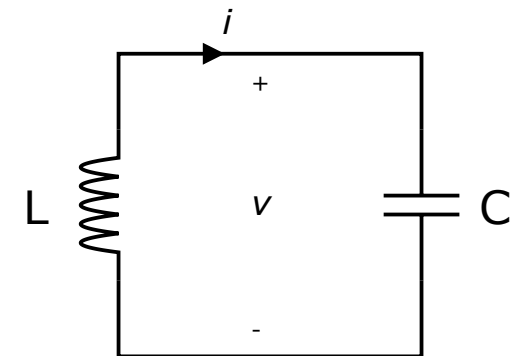
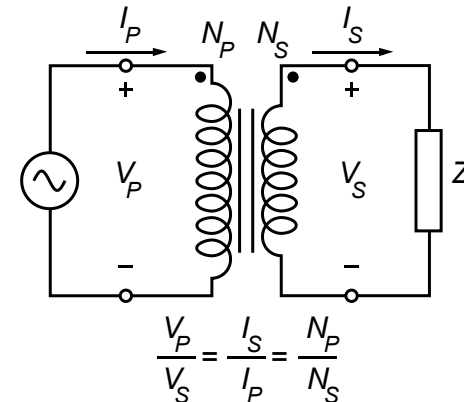
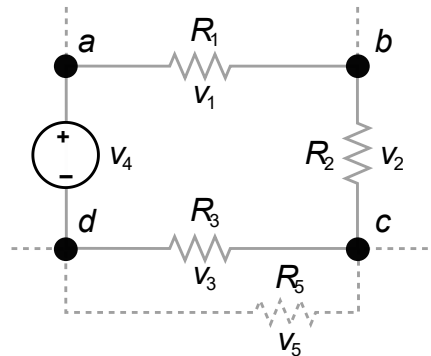
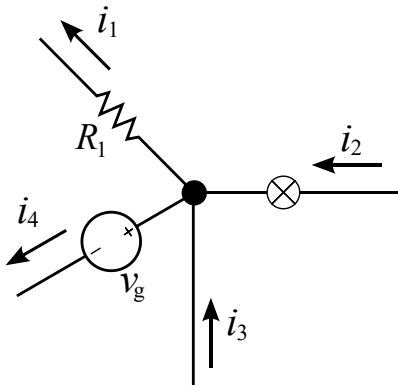


Introduction to Electrical Engineering



Overview

1. Physical Principles and Basic Concepts
2. Direct Current Circuits
3. Electric and Magnetic Fields
4. Alternating Current Technology
- 5. Wrap-Up and Outlook**

Wrap-Up and Outlook

- Summary using Applications
- Outlook
 - Semiconductor Devices and Electronics
 - Energy Systems
 - Information Systems
 - Control Systems
 - Embedded Systems

Fuse and Circuit Breaker

protect circuits from excess current, which is usually caused by:

- overload (more devices than the circuit can handle)
- short circuit (very low resistance path)

typically 16 A for power outlets:

- maximum continuous power (assuming 230 V): $P = V \cdot I = 230 \text{ V} \cdot 16 \text{ A} = 3680 \text{ W}$
- this is enough to power several devices at once, like: 1–2 kettles, 1 microwave, some small appliances together
- Beyond 16 A, the circuit breaker trips to prevent overheating and fire risk.

Thermistor

temperature-dependent resistance

applied in electronic thermometers
and thermostats (control device):

By measuring the resistance, the
corresponding temperature can be
inferred.

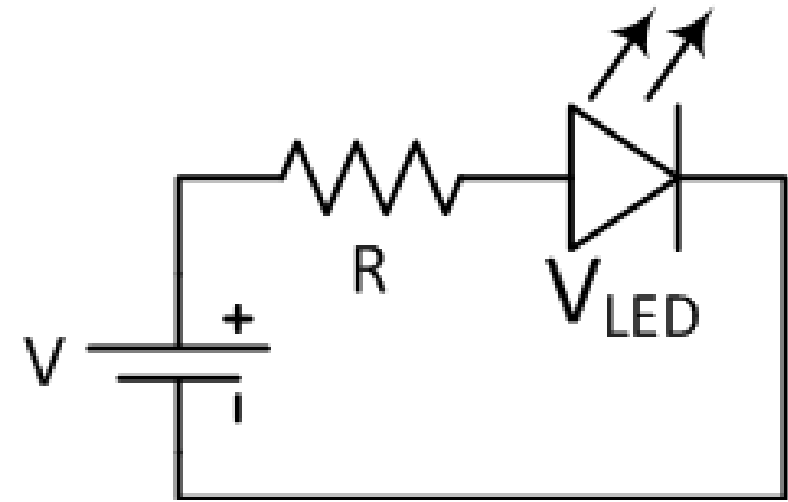


LED and Resistor Circuit

DC circuit (LED is diode)

resistor limits current through LED

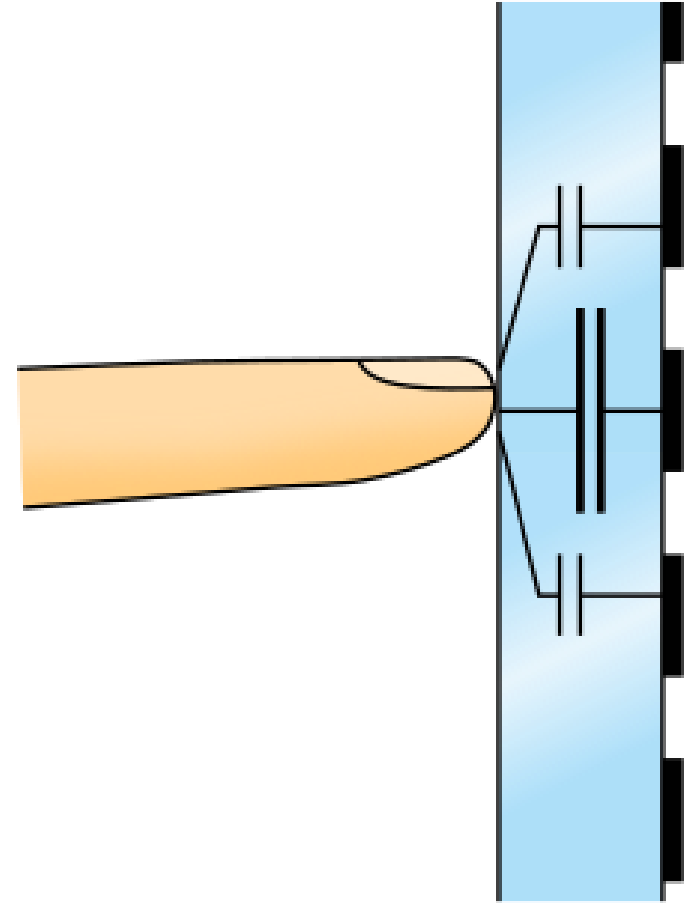
→ preventing damage



Capacitive Touchscreen

usage of electric fields for sensing

- finger changes local capacitance between electrodes
- variation is detected and processed to determine touch position



Inductive Sensor

enables detection of metallic objects without physical contact

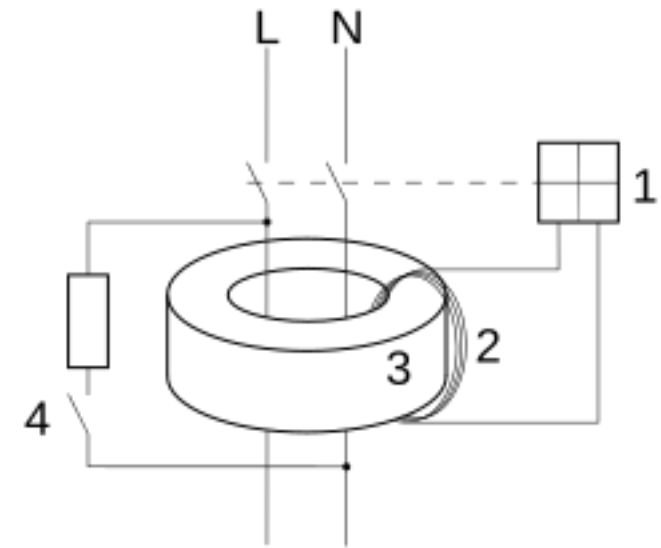
operation based on electromagnetic induction:
When a metal object enters the magnetic field produced by a coil, eddy currents alter the coil's impedance.



Residual Current Device (RCD)

circuit for fault current measurement (30 mA):

- Both conductors pass through a toroidal magnetic core.
- The equally large but opposing currents normally produce magnetic fields that cancel each other, resulting in no net magnetic flux.
- If a fault occurs (such as a person touching a live part) some current flows to ground (leakage current), creating an imbalance.
- This imbalance produces a net magnetic flux in the core, which induces a voltage in a secondary winding.
- The induced signal activates a trip mechanism that disconnects the circuit within milliseconds, preventing electric shock.



- 1: Electromagnet
- 2: Current transformer secondary winding
- 3: Transformer core
- 4: Test push-button
- L: Line conductor
- N: Neutral conductor

Electric Motors

working principle: convert electrical energy into mechanical motion through electromagnetic interaction

- **AC motor:** rotating magnetic field generated by AC, producing torque without direct electrical contact to the rotor
- **DC motor:** commutator and brushes to reverse rotor current every half revolution, maintaining continuous rotation under DC

AC vs DC in Devices

simple appliances (kettles, irons, heaters) run directly on AC

electronics (laptops, phones, TVs) run on DC internally

- chips and memory require a constant voltage to work properly
- semiconductors (CPU, GPU, LEDs) only allow current in one direction
- DRAM, flash, and other memory need consistent DC to store data
- AC fluctuations could damage circuits or cause malfunction

→ adapters to convert AC to DC

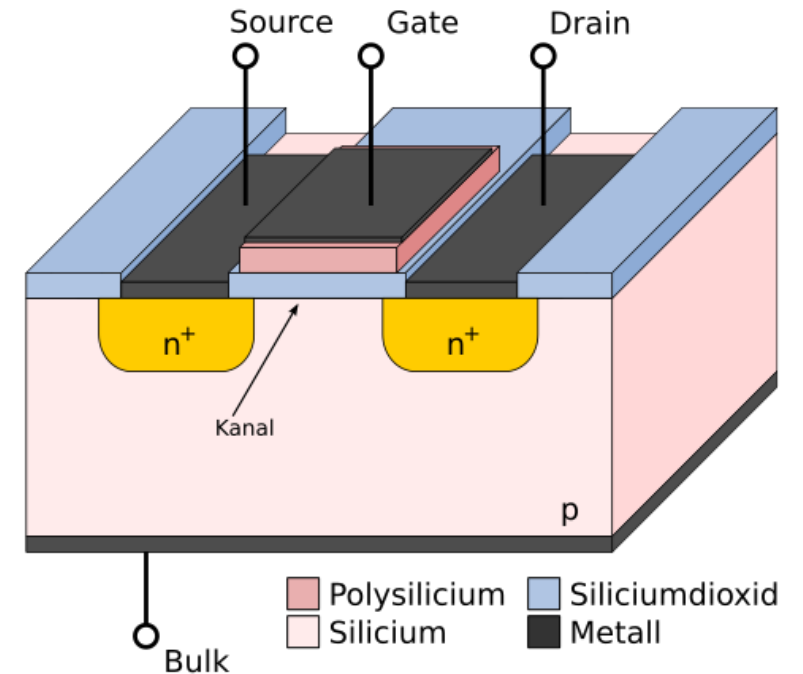
Smartphone Charger



- input: AC from wall outlet (e.g., 230 V)
- diode bridge (4 diodes) converts AC into pulsating DC (rectification)
- capacitor smooths pulsating DC
- switching regulator steps high DC voltage down to required level (e.g., 5 V), using a transistor (switch) and a high-frequency transformer
- output: stable DC voltage to charge battery

Semiconductor Devices and Electronics

- diodes and transistors — switching and amplification
- operational amplifiers — building blocks for analog circuits
- analog circuit design — filters, amplifiers, signal conditioning
- digital electronics — logic gates



Energy Systems

generation, conversion, and distribution of electrical power

- transformers and power converters
- power electronics



Information Systems

transmitting and processing information

signal processing

- time-domain vs. frequency-domain analysis
- Fourier and Laplace transforms

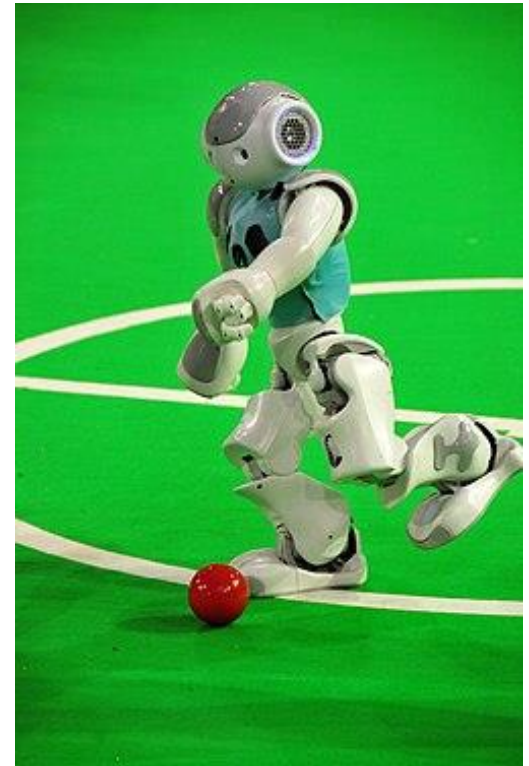
communication systems

- amplitude/frequency modulation (AM/FM)
- digital modulation (ASK, FSK, PSK)
- noise, bandwidth, and filtering concepts
- communication channels and data transmission



Control Systems

- feedback, stability, and dynamic response
- root locus, Bode, and Nyquist techniques
- PID control and real-world applications (motors, robotics)



Embedded Systems

hardware–software integration

- microcontrollers and digital systems
- sensors, actuators, and real-time control

